

§11.6 - Absolute Convergence and the Ratio and Root Tests

Definition #1 (Absolute Convergence)

Let (a_n) be a sequence.

If

$$\sum |a_n|$$

converges then the series $\sum a_n$ **converges absolutely**.

Definition #2 (Conditional Convergence)

Let (a_n) be a sequence.

If

$$\sum |a_n| = \infty$$

diverges but

$$\sum a_n$$

exists then the series $\sum a_n$ **converges conditionally**.

Example #1

The series

$$\sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{\sqrt{k}}$$

converges conditionally while

$$\sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k^2 + \sqrt{k}}$$

converges absolutely.

We know

$$b_k = \frac{1}{\sqrt{k}}$$

is decreasing to 0 so

$$\Rightarrow \sum_{k=1}^{\infty} (-1)^{k+1} \frac{1}{k^{1/2}}$$

converges by the Alternating Series test.

Now

$$\sum_{k=1}^{\infty} |b_k| = \sum_{k=1}^{\infty} \frac{1}{k^{1/2}} = \infty$$

is a divergent p -series since $p = 1/2 \leq 1$.

$$\Rightarrow \sum_{k=1}^{\infty} (-1)^{k+1} \frac{1}{k^{1/2}}$$

converges conditionally.

Since

$$0 < k^2 < k^2 + k^{1/2}$$

$$0 < \frac{1}{k^2 + k^{1/2}} < \frac{1}{k^2}$$

we know

$$\sum_{k=1}^{\infty} |a_k| = \sum_{k=1}^{\infty} \frac{1}{k^2 + k^{1/2}}$$

converges by the Basic Comparison Test
because

$$\sum_{k=1}^{\infty} \frac{1}{k^2}$$

is a convergent p -series since $p = 2 > 1$. Hence

$$\sum_{k=1}^{\infty} (-1)^{k+1} \frac{1}{k^2 + k^{1/2}}$$

converges absolutely.

Theorem #1

Suppose $\sum a_n$ converges absolutely.

Then $\sum a_n$ converges.

Notice that

$$0 \leq |a_n| - a_n \leq 2|a_n|$$

so if $\sum 2|a_n|$ converges, so must $\sum (|a_n| - a_n)$ by the BCT.

Then

$$\sum a_n = \sum |a_n| - \sum (|a_n| - a_n)$$

tells us $\sum a_n$ must converge because it is a difference of two convergent series.

Example #2

We showed in §11.5:

$$\ln(2) = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \dots$$

$$\frac{1}{2} \ln(2) = \left(1 - \frac{1}{2}\right) - \frac{1}{4} + \left(\frac{1}{3} - \frac{1}{6}\right) - \frac{1}{8} + \left(\frac{1}{5} - \frac{1}{10}\right) - \frac{1}{12} + \dots$$

Theorem #2 (Riemann's Rearrangement Theorem)

Let $x \in \mathbf{R}$.

Suppose

$$S = \sum_{k=1}^{\infty} a_k$$

converges conditionally.

Then we can rearrange the terms in the series so that the sum of the reordered terms converges to x .

Warning

The reordering of the terms in an absolutely convergent series does not change the value of the sum.

Definition (Geometric Series)

Let $x \in \mathbf{R}$.

The **geometric series**

$$\sum_{k=0}^{\infty} x^k = \frac{1}{1-x}$$

exists only for $|x| < 1$.

Theorem #3 (Root Test)

Let (a_n) be a sequence and define

$$L = \lim_{n \rightarrow \infty} |a_n|^{1/n}.$$

Then

1. $L < 1$ implies $\sum a_n$ converges absolutely
2. $L > 1$ implies $\sum a_n$ diverges
3. $L = 1$ is inconclusive so use another test.

Idea:

$$|a_n|^{1/n} \rightarrow L \Rightarrow |a_n| \approx L^n$$

and so

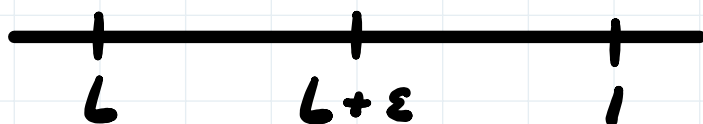
$$\sum |a_n| \approx \sum L^n.$$

Our series "looks like" a geometric sum for n -large.

To prove this note for $L \in (0, 1)$ and

$$\varepsilon = \frac{1-L}{2}$$

half the distance from 1 to L ,



$$\begin{aligned} L + \varepsilon &= L + \frac{1}{2} - \frac{L}{2} \\ &= \frac{L}{2} + \frac{1}{2} \\ &< 1 \end{aligned}$$

since $L < 1$. With $|a_n|^{1/n} \rightarrow L$ choose n -large where

$$||a_n|^{1/n} - L| < \varepsilon = \frac{1-L}{2}$$

$$\Rightarrow |a_n|^{1/n} - L < \varepsilon$$

$$\Rightarrow |a_n|^{1/n} < L + \varepsilon < 1$$

$$\Rightarrow |a_n| < (L + \varepsilon)^n.$$

By the Basic Comparison Test

$$\sum a_n$$

converges absolutely since

$$\sum (L + \varepsilon)^n$$

is a convergent geometric series since $L + \varepsilon < 1$.

For $L > 1$ we have (a_n) not tending toward 0 and so

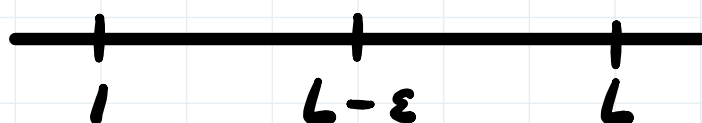
$$\sum a_n$$

diverges by the Basic Divergence Test (BDT).

Details: If $L > 1$ choose

$$\varepsilon = \frac{L-1}{2}$$

and so $1 < L - \varepsilon$.



Then for n -large

$$||a_n|^{1/n} - L| < \varepsilon = \frac{L-1}{2}$$

$$\Rightarrow -\varepsilon < |a_n|^{1/n} - L < \varepsilon$$

$$\Rightarrow 1 < L - \varepsilon < |a_n|^{1/n}$$

$$\Rightarrow 1 < (L - \varepsilon)^n < |a_n|.$$

Clearly, (a_n) is not near 0 for n -large and so the series diverges by the BDT.

Example #3

Determine the convergence of

$$\sum \left(\frac{4n^2 - 3}{7n^2 + 6} \right)^n .$$

Let

$$a_n = \left(\frac{4n^2 - 3}{7n^2 + 6} \right)^n$$

then

$$|a_n|^{1/n} = \left| \frac{4n^2 - 3}{7n^2 + 6} \right| \rightarrow \frac{4}{7} < 1$$

Hence

$$\sum \left(\frac{4n^2 - 3}{7n^2 + 6} \right)^n$$

converges absolutely by the Root Test.

Example #4

Determine the convergence of

$$\sum \frac{2^n}{n^{10}}.$$

We have

$$a_n = \frac{2^n}{n^{10}}$$

giving

$$|a_n|^{1/n} = \frac{2}{n^{10/n}} = \frac{2}{(n^{1/n})^{10}}.$$

What does $n^{1/n}$ do as $n \rightarrow \infty$?

$$n^{1/n} = e^{\log(n^{1/n})} = e^{\frac{1}{n} \log(n)}$$

and for $n > 4$

$$0 < \frac{1}{n} \log(n) \leq \frac{\sqrt{n}}{n} = \frac{1}{\sqrt{n}}.$$

Thus

$$\frac{1}{n} \log(n) \rightarrow 0 \Rightarrow e^{\frac{1}{n} \log(n)} \rightarrow e^0 = 1$$

since e^x is continuous on $\mathbb{R}$:

$$n^{1/n} \rightarrow 1.$$

This shows

$$|a_n|^{1/n} = \frac{2}{(n^{1/n})^{10}} \rightarrow 2 = L > 1$$

and so

$$\sum \frac{2^n}{n^{10}}$$

diverges by the Root Test.

Theorem #4 (Useful Limits)

1. $n^{1/n} \rightarrow 1$

2. $|a|^{1/n} \rightarrow 1$ for $a \neq 0$

3. $\left(1 + \frac{x}{n}\right)^n \rightarrow e^x$

e is such that

$$1 = \lim_{h \rightarrow 0} \frac{e^h - 1}{h}$$

but we can also define it by

$$\lim_{h \rightarrow 0} (1 + h)^{1/h} = e.$$

With $n = \frac{1}{h}$

$$e = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = \lim_{h \rightarrow 0^+} (1 + h)^{1/h}.$$

For $n = xm$,

$$m \rightarrow \begin{cases} \infty, & x > 0 \\ -\infty, & x < 0 \end{cases}$$

as $n \rightarrow \infty$.

If $x > 0$,

$$\begin{aligned} \lim_{n \rightarrow \infty} \left(1 + \frac{x}{n}\right)^n &= \lim_{m \rightarrow \infty} \left(1 + \frac{1}{m}\right)^{xm} \\ &= \left(\lim_{m \rightarrow \infty} \left(1 + \frac{1}{m}\right)^m \right)^x \\ &= e^x. \end{aligned}$$

If $x < 0$,

$$\lim_{n \rightarrow \infty} \left(1 + \frac{x}{n}\right)^n = \left(\lim_{m \rightarrow -\infty} \left(1 + \frac{1}{m}\right)^m\right)^x = e^x$$

since e can be expressed by

$$e = \lim_{m \rightarrow -\infty} \left(1 + \frac{1}{m}\right)^m = \lim_{h \rightarrow 0^-} (1+h)^{1/h}.$$

We have shown

$$\left(1 + \frac{x}{n}\right)^n \rightarrow e^x$$

for all $x \in \mathbb{R}$.

Definition #3 (Factorial)

Let $n \in \mathbf{N}$.

Then

$$\begin{aligned} n! &= n \times (n - 1) \times (n - 2) \times \cdots \times 3 \times 2 \times 1 \\ &= n(n - 1)! \end{aligned}$$

defines the **n th factorial**.

We have

$$1! = 1$$

$$2! = 2 \cdot 1 = 2$$

$$3! = 3 \cdot 2 \cdot 1 = 6$$

$$4! = 4 \cdot 6 = 24$$

$$5! = 5 \cdot 24 = 120$$

and we define $0! = 1$.

Theorem #5 (Ratio Test)

Let (a_n) be sequence of non-zero numbers and define

$$L = \lim_{n \rightarrow \infty} \frac{|a_{n+1}|}{|a_n|}.$$

Then

1. $L < 1$ implies $\sum a_n$ converges absolutely
2. $L > 1$ implies $\sum a_n$ diverges
3. $L = 1$ is inconclusive so use another test.

Idea: For n -large,

$$\begin{aligned} \left| \frac{a_{n+1}}{a_n} \right| &\approx L \Rightarrow |a_{n+1}| \approx L|a_n| \\ \Rightarrow |a_{n+2}| &\approx L|a_{n+1}| \approx L^2|a_n| \\ \Rightarrow \sum_{k=n}^{\infty} |a_k| &\approx |a_n|(1 + L + L^2 + \dots) \\ &= |a_n| \sum_{k=0}^{\infty} L^k. \end{aligned}$$

If $L < 1$ we have convergence and if $L > 1$ we have divergence.

For the proof we can modify our Root Test argument: If $L < 1$ choose n -large such that

$$\left| \left| \frac{a_{n+1}}{a_n} \right| - L \right| < \varepsilon = \frac{1-L}{2}$$

so

$$|a_{n+1}| < (L + \varepsilon)|a_n| \Rightarrow |a_{n+2}| < (L + \varepsilon)^2|a_n|$$

and therefore

$$\sum_{k=n}^{\infty} |a_k| \leq |a_n| \sum_{k=0}^{\infty} (L + \varepsilon)^k.$$

With $|L + \varepsilon| < 1$, we have convergence of $\sum |a_k|$ by the BCT.

Example #5

Determine the convergence of

$$\sum \frac{n^2}{n!}.$$

Here

$$a_n = \frac{n^2}{n!}$$

and

$$(n+1)! = (n+1)n!$$

gives

$$\begin{aligned} a_{n+1} &= \frac{(n+1)^2}{(n+1)!} \\ &= \frac{(n+1)^2}{(n+1)n!} \\ &= \frac{n+1}{n!}. \end{aligned}$$

Next,

$$\begin{aligned} \frac{a_{n+1}}{a_n} &= \frac{\frac{n+1}{n!}}{\frac{n^2}{n!}} \\ &= \frac{n+1}{n^2} \rightarrow 0. \end{aligned}$$

and with $L = 0 < 1$,

$$\sum \frac{n^2}{n!}$$

converges by the Ratio Test.

Warning

Sums of powers of n will often not work with the Root or Ratio Test.

For

$$\sum \frac{1}{n} \quad \text{;} \quad \sum \frac{1}{n^2}$$

the Root Test fails

$$\left(\frac{1}{n}\right)^{1/n} \rightarrow 1 \quad \text{;} \quad \left(\frac{1}{n^2}\right)^{1/n} \rightarrow 1$$

and the Ratio Test fails

$$\frac{\frac{1}{n+1}}{\frac{1}{n}} = \frac{n}{n+1} \rightarrow 1 \quad \text{;} \quad \frac{\frac{1}{(n+1)^2}}{\frac{1}{n^2}} = \frac{n^2}{(n+1)^2} \rightarrow 1.$$

Example #6

Determine the convergence of

$$\sum \frac{n^n}{n!}.$$

Here

$$a_n = \frac{n^n}{n!}$$

so

$$\begin{aligned} a_{n+1} &= \frac{(n+1)^{n+1}}{(n+1)!} \\ &= \frac{(n+1)(n+1)^n}{(n+1)n!} \\ &= \frac{(n+1)^n}{n!}. \end{aligned}$$

Notice

$$\begin{aligned} \frac{a_{n+1}}{a_n} &= \frac{\frac{(n+1)^n}{n!}}{\frac{n^n}{n!}} \\ &= \left(\frac{n+1}{n} \right)^n \\ &= \left(1 + \frac{1}{n} \right)^n \rightarrow e \end{aligned}$$

and $e > 1$. Thus

$$\sum \frac{n^n}{n!}$$

diverges by the Ratio Test.

Remark

- Root Test: good for powers of n
- Ratio Test: good for $n!$ or recursive terms
- Root Test is stronger than the Ratio Test, it works for more series. (Take Math 104)